



Measurement 12: Noise as the Signal

Measurement Exercise for Physicist-Engineer Students

Abstract. When measuring different quantities, we are usually interested in the expected value of the quantity under test and consider fluctuations around the expected value as a disturbing factor. In many cases, however, the noise of a physical quantity carries more information about the system than the expected value itself. On this measurement occasion, different noise phenomena are investigated using a simple data acquisition board and the low-noise voltage amplifier. In the first session, we will learn the basics of signal processing and then determine the value of the Boltzmann constant by measuring the thermal noise of resistors. In the second measurement session, the value of the electron charge is determined from the noise of a semiconductor diode, and a system with a $1/f$ -type noise spectrum is also investigated.

Warning. For this measurement, a sensitive voltage amplifier and a data acquisition card are used. Before applying a voltage to these, please check the maximum signals that the instruments can withstand and never exceed these absolute maximum ratings. Some of the instruments are powered by standard 230 V household voltage. Before powering them up, carefully inspect both the instruments and their power cables to ensure that the housing and insulation are not damaged.

If you find yourself uncertain about any aspect, do not hesitate to seek help from the responsible tutors present in the laboratory. Always respect their warnings and instructions.

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Remark. This measurement description relies heavily on the noise sections of the Measurement Techniques course (BMETE11AP65), which you may want to review in preparation for the measurement.

Theoretical summary

1 Noise basics

Noise is commonly considered as a disturbing factor. In traditional electrical engineering, it is indeed a challenge to reduce the inherent noise of the devices as much as possible, and in everyday life we talk about noisy neighbors, noisy phone calls, etc. However, in metrology, noise can also serve as a useful source of information. For example, the thermal noise

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of a resistor can be considered as a temperature standard that converts the temperature measurement into voltage and current measurements via the Boltzmann constant, and without any material-dependent parameters. These considerations show that the noise character of a signal does not depend on whether it is disturbing or useful, but a more rigorous approach is required in the language of signal analysis.

From this more scientific point of view, noise is treated as a special type of signal, with a broad and rather continuous spectrum of contributions from all frequencies. For such a signal, the mean squared deviation within a small Δf bandwidth around the central frequency f_0 scales with the bandwidth due to the superposition of the different frequency components, and the proportionality factor is the *spectral density of noise*, or the *noise power spectrum density* (PSD). For instance a fluctuating $I(t)$ current can be decomposed to its $\langle I(t) \rangle$ mean value and the $\Delta I(t)$ temporal fluctuation (noise) around the mean value (Fig. I.1.1):

$$I(t) = \langle I(t) \rangle + \Delta I(t). \quad (\text{I.1.1})$$

If we measure the $\langle (\Delta I(t|f_0, \Delta f))^2 \rangle$ mean squared deviation of this fluctuating signal with a band-pass filter using a sufficiently narrow Δf bandwidth around the f_0 central frequency (Fig. I.1.2), the results scales with Δf :

$$\langle (\Delta I(t|f_0, \Delta f))^2 \rangle = S_I(f_0) \cdot \Delta f. \quad (\text{I.1.2})$$

Here, the $S_I(f)$ proportionality factor is the spectral density of the current noise, which is considered as an *experimental definition* of the current noise spectrum. This definition also makes it clear that a signal with a distinctive frequency component, i.e. a sharp peak in the spectrum (see Fig. I.1.3(b)), is not noise because the signal's mean squared deviation does not scale with the Δf bandwidth, i.e. the continuous nature of the spectrum is an essential property of a noise signal (see Fig. I.1.3(a)). For such a continuous signal S_I considered as the *density* of the noise, and the mean squared deviation over a much broader frequency interval is obtained by integration:

$$\langle (\Delta I(t))^2 \rangle = \int_{f_1}^{f_2} S_I(f) df. \quad (\text{I.1.3})$$

From a theoretician's point of view, the description of noise is usually approached through correlation functions. Fortunately, these approaches are equivalent: according to the so-called Wiener-Khinchin theorem, the above defined $S_I(f)$ spectral density is twice the Fourier-transform of the current-current correlation function, $C_I(t_2 - t_1) = \langle \Delta I(t_1) \cdot \Delta I(t_2) \rangle$. Imagine a signal, where the fluctuations at different $t_1 \neq t_2$ times are completely independent of each other. This can be mathematically described by the $C_I(t_2 - t_1) = C_0 \cdot \delta(t_2 - t_1)$ Dirac delta correlation function whose Fourier transform gives a constant, frequency-independent noise, called white noise: $S_I(f) = 2C_0$. We can also consider a signal, where the fluctuations are only independent of each other, if the $\Delta t = t_2 - t_1$ time difference is larger than a certain τ correlation time. This case is well described by a $C_I(\Delta t) = C_0 \exp(-|\Delta t|/\tau)$ exponential correlation function, which yields the

$$S_I(f) = \frac{4C_0\tau}{1 + (2\pi f)^2/\tau^2} \quad (\text{I.1.4})$$

noise spectrum after Fourier transformation. Note, that this is a white noise at frequencies well below the inverse correlation time ($2\pi f \ll \tau^{-1}$), whereas the noise spectrum decays with $1/f^2$ in the high frequency limit ($2\pi f \gg \tau^{-1}$).

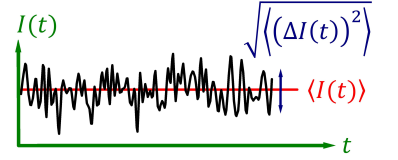


Figure I.1.1. A noisy $I(t)$ signal can be considered as the sum of its $\langle I(t) \rangle$ mean value and its $\Delta I(t)$ fluctuation around the mean value, $I(t) = \langle I(t) \rangle + \Delta I(t)$. The fluctuations can be characterized by the $\langle (\Delta I(t))^2 \rangle$ mean squared deviation of the signal, or its square root, the standard deviation of the signal.

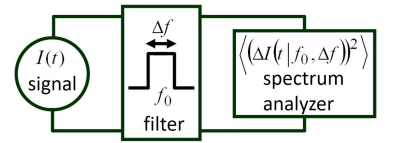


Figure I.1.2. Experimental definition of noise: the mean squared deviation of an $I(t)$ current signal within a Δf band around the f_0 central frequency normalized to Δf gives us the S_I spectral density of the current noise.

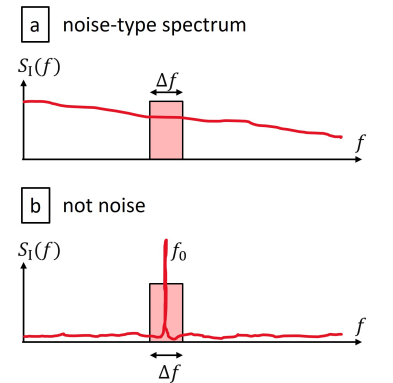


Figure I.1.3. (a) A noise-type spectrum is continuous, and therefore the mean squared deviation of the various frequency components adds up. From this reason the mean squared deviation is indeed proportional to the Δf bandwidth for a narrow enough band. (b) A spectrum with a distinct frequency component (even if it is a disturbing pick-up signal) does not satisfy the experimental definition of noise, i.e. it is not a noise-type signal.

The above experimental and theoretical approaches to noise are useful for understanding the nature of noise signals, but none of them provide a practical tool for evaluating the noise spectrum from the measured $I(t)$ time-trace. To this end, we can use another identity, namely that the $S_I(f)$ spectral density of noise is proportional to $\langle |\Delta i(f)|^2 \rangle$, i.e., the square of the absolute value of $\Delta i(f)$, which is the Fourier transform of the $\Delta I(t)$ temporal fluctuation around the average current. More specifically, this relationship reads as follows:

$$S_I(f) = \lim_{T \rightarrow \infty} \frac{2}{T} \left\langle \left| \int_{-T/2}^{T/2} dt \Delta I(t) e^{-i2\pi f t} \right|^2 \right\rangle. \quad (\text{I.1.5})$$

This can be easily evaluated for discrete sample points using the discrete Fourier transform (DFT):

$$S_I(f) = \frac{2\Delta t}{N} \cdot \left\langle \left| \sum_{n=0}^{N-1} \Delta I(n\Delta t) e^{-i2\pi f n\Delta t} \right|^2 \right\rangle, \quad (\text{I.1.6})$$

where N is the number of data points acquired, and Δt is the time difference between the adjacent samples, such that $N \cdot \Delta t = T$ is the duration of the measurement. Usually several time-traces are recorded, the DFT spectrum is evaluated for each of them, and the expectation value in the above formula is calculated by averaging the different spectra. Note, however, that the brute-force calculation of the spectrum by DFT requires $\approx N^2$ operations for N data points. A much more efficient calculation method is offered by the so-called fast Fourier transform (FFT) algorithm, which requires $N \cdot \log_2(N)$ operations. This utilizes some symmetries of the DFT formula, but this simplification only works out for $N = 2^p$ data-points, where p is an integer. This means, that an FFT spectrum analyzer always takes an integer power of 2 data-points from the acquired dataset, e.g. from a time series of 2000 data-points it only uses the first 1024 data-points to calculate the spectrum, and skips the rest of the data.

So far we have considered current noise, whose S_I spectral density has the unit of A^2/Hz . In engineering practice, the square root of the spectral density is also often used to describe the noise of an instrument in the units of $\text{A}/\sqrt{\text{Hz}}$.

Similarly to current noise, voltage noise can be defined as

$$\langle (\Delta V(t|f_0, \Delta f))^2 \rangle = S_V(f_0) \Delta f. \quad (\text{I.1.7})$$

For a simple resistor the ΔV voltage fluctuation is calculated by Ohm's law from the ΔI current fluctuation: $\Delta V = R \cdot \Delta I$. From this the $S_V = R^2 \cdot S_I$ conversion follows for the spectral densities of voltage and current noise. For a nonlinear device, such as a diode, however, the $\Delta I \leftrightarrow \Delta V$ conversion depends on the operation point, i.e. the V_0 voltage at which the device is operated. For small enough variations around the operation point $\Delta V = R_d \cdot \Delta I$ holds (see Fig. I.1.4), where $R_d = \left. \frac{dV}{dI} \right|_{V_0}$ is the differential resistance at V_0 . From this the $S_V = R_d^2 \cdot S_I$ conversion follows for the spectral densities of voltage and current noise.

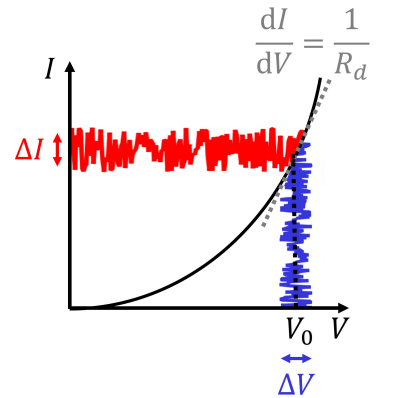


Figure I.1.4. Illustration of the differential resistance and conversion of voltage and current fluctuations for a nonlinear device.

2 Noise types

2.1 Noise of gun shots and shot noise of electrons

The concept of noise can be illustrated by a classic example, let's look at what happens when a rifle is fired randomly, with the firing times being completely independent of each other. If the average time between adjacent shots is τ , then the average number of shots

over time Δt is $\langle N \rangle = \Delta t / \tau$. However, the actual N number of shots within Δt time will obviously fluctuate around the average value. To determine the mean squared deviation of N , it is worth calculating the probability $P_N(\Delta t)$, i.e. the probability that N shots will be fired in Δt . It can be shown that independent shots are described by the so-called Poisson distribution according to the

$$P_N(\Delta t) = \frac{(\Delta t)^N}{\tau^N N!} e^{-\Delta t / \tau} \quad (\text{I.2.1})$$

formula. A special property of the Poisson distribution is that the mean squared deviation of N is equal to its mean value value, i.e.

$$\langle (\Delta N)^2 \rangle = \langle N \rangle = \frac{\Delta t}{\tau}. \quad (\text{I.2.2})$$

The above reasoning can also be applied to electrons if it is true that electrons pass from one electrode to another randomly and independently of each other. Suppose that our measuring system can measure the electric current with a time resolution of Δt . During a sampling interval of width Δt we detect a current $I = Ne / \Delta t$ where the distribution of the number of electrons N passing through the electrodes during time Δt is given by the Poisson distribution above. Thus the mean value of the measured current is $\langle I \rangle = \langle N \rangle e / \Delta t$, while the mean squared deviation of the current is $\langle (\Delta I)^2 \rangle = \langle (\Delta N)^2 \rangle e^2 / (\Delta t)^2 = \langle N \rangle e^2 / (\Delta t)^2 = \langle I \rangle e / \Delta t$. Note, that in this scheme we do not measure the current continuously, but we choose Δt long time intervals, along which we measure the actual N number of electron incidences, i.e. we study a quantity which is integrated over this time interval. The mean value and the mean squared deviation of the current is calculated for the set of N values measured for a large number of subsequent Δt wide time intervals. With this method we can only resolve the current fluctuations on time-scales longer than Δt , which sets a corresponding upper frequency limit $f_{\max} = 1 / (2\Delta t)$ for our analysis. Accordingly the above calculated $\langle (\Delta I)^2 \rangle$ mean squared current deviation is the integral of the current noise's spectral density up to this $f_{\max}$ cutoff frequency:

$$\langle (\Delta I)^2 \rangle = \langle I \rangle e / \Delta t = \int_0^{f_{\max}=1/(2\Delta t)} S_I(f) df. \quad (\text{I.2.3})$$

This relation is satisfied for any choice of Δt if the spectral density of the current noise is frequency independent with a value of

$$S_I = 2e \langle I \rangle. \quad (\text{I.2.4})$$

By analogy with the rifle shots, the current noise caused by the discrete charge of electrons is also called *shot noise*. It is important to emphasize again that, according to the above formula, the shot noise of electrons is a frequency independent white noise.

Supplementary Information. It is noted, that the noise formula derived above describes a special case of electron shot noise, the so-called Poisson noise, which refers to the detection of independently emitted electrons. According to Pauli's principle, known from quantum mechanics, no two electrons can be in the same state, i.e. no two electrons in exactly the same state can be detected at a given time. Due to this property, the electrons of a macroscopic conductor do not arrive at the current meter independently of each other, but rather *in sequence*. This violation of the independence condition also means that the above electron shot noise formula is not

valid for a general case (not even for a simple resistor), rather a more sophisticated analysis is required to calculate the noise formula for a particular experimental arrangement. These rich properties of electron shot noise are broadly utilized in the field of nanophysics to investigate quantum conductance channels, to measure the double electron charge of elementary carriers in superconductors, to distinguish classical or quantum mechanical chaotic behavior, or to recognize the so-called bosonic or fermionic nature of the investigated particles. These issues will be covered later by specialized nanophysics courses, and a short summary is available [in this review paper by C. Beenakker and C. Schöenberger](#).

Noting the above supplementary information, now we turn to systems, where the independence condition is accurately satisfied. This is generally the case, if some kind of barrier is placed in the path of the electrons, through which only a small fraction of the electrons can pass at random.

The first measurement of shot noise was carried out by Walter Schottky in 1918: his famous experiment investigated the noise of the anode current of a vacuum diode. The construction of the vacuum diode is illustrated in Fig. I.2.1. Electrons randomly emerging from a heated cathode are driven by a voltage applied between the cathode and the anode to the anode, where a current is detected. The vacuum diode is an ideal tool for studying shot noise, since the electrons are indeed emitted randomly and independently of each other. This noise measurement method served as an alternative to Millikan's famous experiment to determine the electron charge.

Electron shot noise described by Eq. I.2.4 can also be seen in modern electronic devices. For example, a semiconductor diode includes a barrier-like interface between different types of semiconductor layers, where the electrons pass randomly and independently with a low transmission probability once the diode is operated at sufficiently low current. This is utilized in the measurement exercise to determine the charge of the electron based on a combined measurement of the expected value of the current and its fluctuation.

For the sake of fairness, it is worth noting that historically it has been a very important task to determine the charge of an electron, but in 2019 the new SI system has fixed the value of the electron charge, so it is no longer a measurable quantity but a fixed natural constant. Suppose we now put ourselves back into the context where the electron charge was a quantity to be measured, and we test how accurately we can determine the electron charge under simple laboratory conditions by measuring the shot noise of a diode.

2.2 Thermal noise

The shot noise described above is a non-equilibrium noise that is only experienced when current is flowing through the circuit element under test. However, noise can also be experienced at equilibrium simply due to thermal fluctuations of the electrons. Understanding *thermal noise* requires more theoretical background (for a detailed derivation, see the handouts of the Measurement Techniques course), but the phenomenon itself can be described by a very simple formula: for a circuit element with an electrical resistance R

$$S_V = 4k_B T \cdot R \quad (\text{I.2.5})$$

voltage noise density can be measured regardless of the material properties of the physical system that provides the resistance R . Thermal noise is also white noise, i.e. the noise density does not depend on frequency. Using this phenomenon, the Boltzmann constant can be determined from the measurement of the voltage noise, given the temperature T

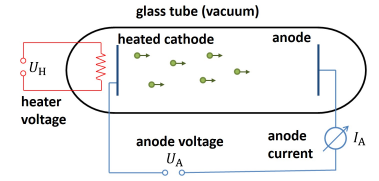


Figure I.2.1. Scheme of the shot noise measurement in a vacuum diode. The electrons are emitted randomly and independently of each other from a heated cathode. Thank to the anode voltage they arrive to the anode, where the anode current exhibits $S_I = 2eI$ current noise power due to the discrete electron incidences.

and the resistance R . Again, it is noted that since 2019 the new SI system considers the Boltzmann constant as a fixed natural constant, which is used to convert temperature measurement to the measurement of other quantities. Here we will check how accurately we can reproduce this natural constant by simply measuring the noise of a resistor. This is an extremely simple scheme: we simply measure the voltage fluctuations on a resistor without applying any driving (see Fig. I.2.2).

2.3 1/f-type noise

Besides thermal noise and shot noise, it is worth discussing the so-called 1/f-type noise, which is named after the typical $1/f^\alpha$ frequency dependence of the noise density ($\alpha \approx 1$). 1/f-type noise typically dominates low frequency measurements, while at higher frequencies thermal noise and, in some devices, shot noise are the most important noise sources.

In electronic devices the 1/f-type noise is usually related to resistance fluctuations. Imagine a piece of a metal, where an atom, or a group of atoms fluctuates between two metastable states with a characteristic time of τ (see bottom inset in Fig. I.2.3). This structural fluctuation yields a resistance fluctuation with the same characteristic time. This is well-described by the exponential correlation function of the resistance fluctuations, $C_R(\Delta t) = \langle \Delta R(t) \cdot \Delta R(t + \Delta t) \rangle = C_0 \exp(-|\Delta t|/\tau)$. From this the so-called Lorentzian noise spectrum of Eq. I.1.4 follows, which is white noise at $2\pi f \ll \tau^{-1}$, whereas it decays with $1/f^2$ at $2\pi f \gg \tau^{-1}$. Such spectra of individual fluctuators is exemplified by red curves in Fig. I.2.3. If our resistor includes many of such fluctuators (see top inset in Fig. I.2.3), all having different τ_i fluctuation time constants, the Lorentzian noise spectra of the different fluctuators adds up yielding an overall noise spectrum with much shallower frequency dependence than $1/f^2$ (see black curve in Fig. I.2.3). This is the so-called 1/f-type noise, which does not necessarily has exactly $1/f$ scaling, rather $S_I(f) \sim 1/f^\gamma$, where γ is usually close to one, and definitely much smaller than two.

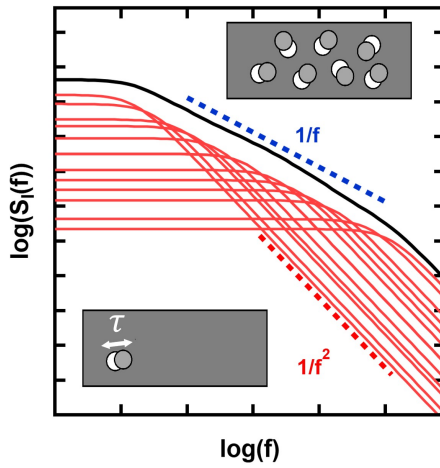


Figure I.2.3. Demonstration of the 1/f-type noise as a resistance fluctuation. The bottom inset illustrates a piece of a metal with a single structural fluctuator with τ characteristic time. This yields a Lorentzian spectrum, like the red curves. For many fluctuators with different time-constants the Lorentzian spectra add up, yielding a spectrum with shallower frequency dependence (black curve).

The above discussed 1/f-type noise is essentially related to resistance fluctuations, but in a measurement rather current noise or voltage noise is measured. In the latter case – due to Ohm’s law – the resistance fluctuation converts to voltage fluctuation as $\Delta V = I \cdot \Delta R$,

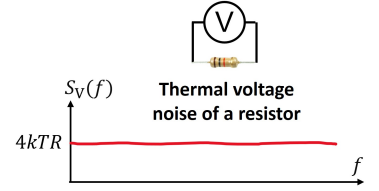


Figure I.2.2. Scheme of the thermal noise measurement: the time-resolved voltage fluctuations are measured on a simple resistor without driving. Note, that classically this scheme would yield zero voltage, but instead we observe the thermal voltage fluctuations, which yield a frequency independent white noise spectrum in the frequency domain.

from which the

$$S_V(f_0) = \langle (\Delta V(t|f_0, \Delta f))^2 \rangle / \Delta f = I^2 \cdot \langle (\Delta R(t|f_0, \Delta f))^2 \rangle / \Delta f = I^2 \cdot S_R \quad (\text{I.2.6})$$

relation follows for the spectral noise densities, i.e. the spectral density of the related voltage noise scales with the driving current squared.

3 Consequences of Discrete Spectrum Analysis, and the Need for an Antialiasing Filter

The evaluation of a Fourier spectrum from a real measurement faces two difficulties, (i) we can only perform the measurement for a finite T time, (ii) we are not measuring continuously, but sampling at discrete Δt time differences. The first constraint yields a finite $\Delta f \approx 1/T$ frequency resolution of the spectrum, while (ii) sets an $f_{\max} = 1/(2\Delta t)$ frequency limit, above which the spectrum cannot be resolved. This is the so-called Nyquist frequency. Now let us assume to have a signal component with f_0 frequency, which is well above the Nyquist frequency. For this $F(t) = A_0 e^{i2\pi f_0 t}$ signal the discrete Fourier transform reads as

$$\text{DFT}(f_k) = \sum_{n=0}^{N-1} A_0 e^{i2\pi f_0 n \Delta t} e^{-i2\pi f_k n \Delta t} \Delta t. \quad (\text{I.3.1})$$

Note that the

$$\text{DFT}(f_k) = \text{DFT}\left(f_k + \frac{m}{\Delta t}\right) \quad (\text{I.3.2})$$

relation is satisfied due to the discrete nature of the spectrum, where m is an integer. On the other hand $\text{DFT}(f_k) = \text{DFT}^*(-f_k)$ is also satisfied, from which

$$|\text{DFT}(f_k)| = |\text{DFT}(-f_k)| \quad (\text{I.3.3})$$

follows. Thus, it can be seen that any signal with a $f > f_{\max}$ circular frequency is seen as if it were a signal in the $[0, f_{\max} = 1/(2\Delta t)]$ domain with its own amplitude A_0 . In the time-domain this so-called aliasing phenomenon is illustrated by Fig. I.3.1. The red signal is the real high-frequency signal, which cannot be resolved due to the too large Δt sampling interval. In this case the lower frequency gray signal appears as a virtual signal in the measurement. From a noise measurements point of view, similar phenomenon is illustrated in Fig. I.3.2 in the frequency domain. The blue spectrum is to be measured, but its higher frequency part extends above the Nyquist frequency of the measurement (right black line). The $f > f_{\max}$ frequency components fold back to the $[0, f_{\max} = 1/(2\Delta t)]$ frequency interval of our analysis due to the aliasing phenomenon, and add up to a measured spectrum (black spectrum), which significantly grows above the true blue spectrum of the signal. This artifact is eliminated, if we apply a low-pass filter (gray dotted line), which cuts off all the signal components right below the Nyquist frequency.

As a specific example, in our measurement we will use 200 kS/s sampling rate, which yields a Nyquist frequency of 100 kHz. Our low noise amplifier has a cutoff frequency right at 100 kHz. But this is only the frequency point, where the signal power is halved, i.e. the amplifier still transmits some signal above the Nyquist frequency due to the upper tail of its filter characteristic. This is already enough to introduce significant error to the noise measurement. Instead, we will use a dedicated anti-aliasing filter with lower, 30 kHz cutoff, which really filters out all significant signal components above the Nyquist frequency.

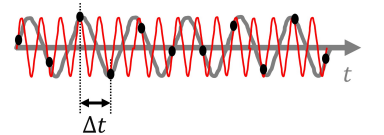


Figure I.3.1. Aliasing phenomenon in the time-domain.

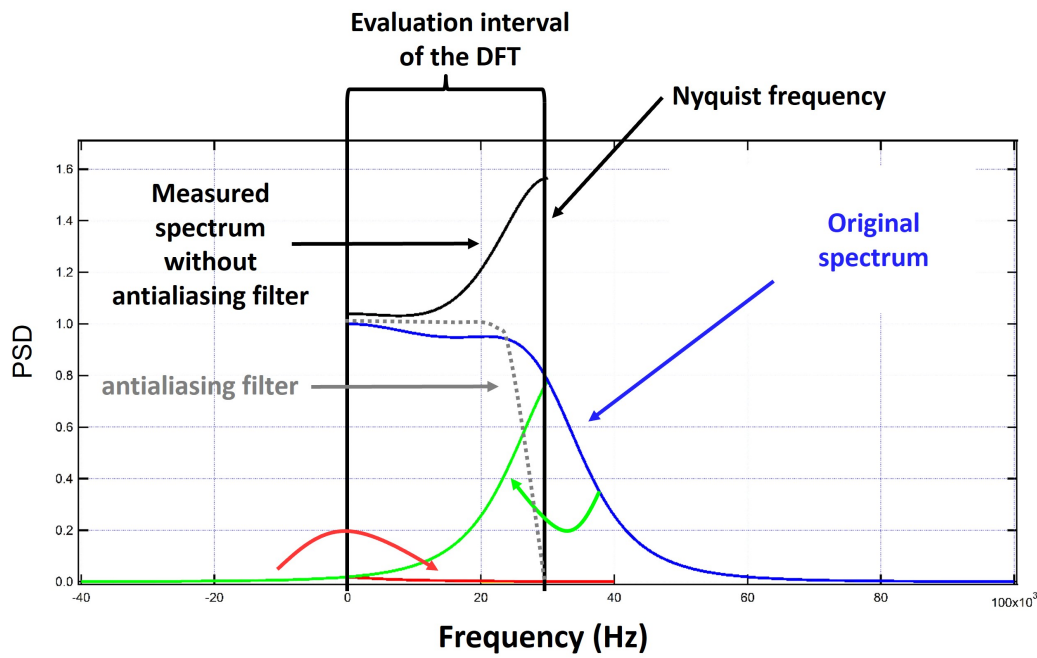


Figure I.3.2. Frequency-domain aliasing phenomenon in a noise measurement. An antialiasing filter (gray dotted curve) eliminates the signal components above the Nyquist frequency, and therefore, eliminates the artifacts due to aliasing.

Measurement tools

II

List of Measurement Tools.

- MyDAQ data acquisition card with holder plate, BNC connectors and USB cable (Fig. II.1.1a)
- FEMTO DLPVA 100-F-S or 101-F-S low noise voltage amplifier with power supply (Fig. II.1.1b)
- Antialiasing filter (Fig. II.1.1c)
- Sample holder box (Fig. II.1.1d)
- Carbon fibre box (Fig. II.1.1e)
- NI ELVIS software (Dynamic Signal Analyzer - Fig. II.3.1, DC Level Output - Fig. II.4.1, Function Generator - Fig. II.5.1, Oscilloscope - Fig. II.5.1, Bode Analyzer - Fig. II.6.1, Digital Multimeter - Fig. II.7.1)
- Screwdriver
- Cables for the myDAQ Digital Multimeter
- Resistors for the thermal noise measurement ($1\text{ k}\Omega$, $3.3\text{ k}\Omega$, $6.8\text{ k}\Omega$ and $10\text{ k}\Omega$)
- Resistor and capacitor for filtering the drive voltage ($330\text{ }\Omega$ and $2200\text{ }\mu\text{F}$)
- Resistors in series for the shot noise measurement ($330\text{ k}\Omega$) and $1/f$ -type noise measurement ($33\text{ k}\Omega$)
- BP 139 bipolar transistor acting as a diode in the shot noise measurement
- BNC T-piece (1 piece), BNC plug – BNC plug adapter (3 pieces), 2 coaxial cables with BNC connectors ($\approx 1\text{ m}$ and $\approx 0.23\text{ m}$), BNC 50Ω termination (1 piece)
- Shielded wires with bare ends (2 pieces, $\approx 13\text{ cm}$), unshielded wires with bare ends (4 pieces, $\approx 13\text{ cm}$), short wires to connect MyDAQ pins (2 pieces)
- Box for the accessories
- Thermometer (common for all measurement pairs)

1 The Measurement Setup

The measurement setup is illustrated in Fig. II.1.1. In this arrangement the carbon fibre box for the $1/f$ -type noise measurement is also included (Fig. II.1.1e). For thermal and shot noise measurements, this element is omitted and the sample holder box (Fig. II.1.1d) is connected directly to the amplifier (Fig. II.1.1b).

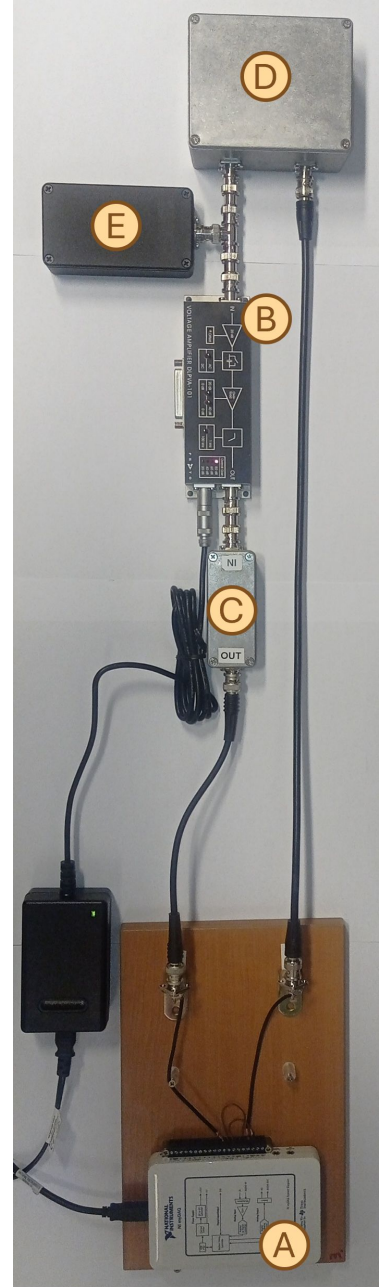


Figure II.1.1. Measurement setup including the myDAQ data acquisition card (a), the FEMTO low noise voltage amplifier (b), the antialiasing filter (c), the sample holder box (d) and the carbon fibre box (e).

2 MyDAQ Data Acquisition Card

For the measurement we use a National Instruments myDAQ data acquisition card. The card is capable of sampling at 200 kS/s and, importantly for the measurement, does not have a built-in anti-aliasing filter. Detailed specifications and descriptions are available in the instrument [datasheet](#), and are briefly summarized in Fig. II.2.1. BNC connectors are attached to one analog input (AI1) and one analog output (AO1) of the instrument for easy connection (see Fig. II.1.1a).

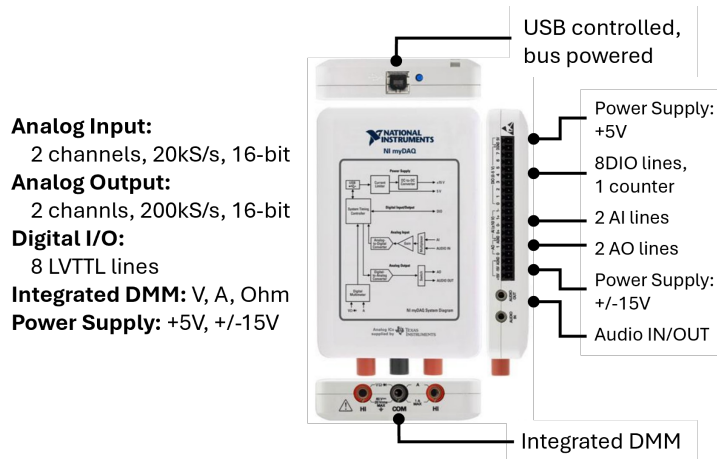


Figure II.2.1. National Instruments myDAQ data acquisition card with its main functionalities and specifications.

3 Using and setting up the Dynamic Signal Analyzer

The data acquisition card is connected to the computer via USB port, where the noise signal is processed using the Dynamic Signal Analyzer in the NI ELVISmx software package. **Please install this software to your computer prior to the measurement!** The Dynamic Signal Analyzer evaluates the Fourier spectrum of the signal measured at the input of the data acquisition card. Its user interface is shown in Fig. II.3.1. The program displays the measured spectrum in decibels, i.e. on a logarithmic scale, while the frequency axis is divided linearly. Below the spectrum, the smaller display shows the measured voltage in the time-domain. The most important setting of the signal analyzer are:

- In the drop-down menu below the display, you can select whether you want to measure the square root of the complex Fourier transform of the signal, i.e. the **Power Spectrum** or the spectral density of noise, i.e. the **power spectrum density (PSD)**. **The latter should be used for the measurements.** The measurement program gives the amplitude of the noise density in $\text{dBVrms}^2/\text{Hz}$. These should be converted to the practical unit of $\text{V}/\sqrt{\text{Hz}}$ during the evaluation.
- In the **Input Settings** menu you can set the sampling channel and the sampling voltage range. (AI1 and $\pm 10\text{V}$ is a proper setting for the entire measurement).
- In the **FFT Settings** menu, you can set the maximum sampling frequency (up to 100 kHz), the number of frequency points (Resolution) and the window function to use. For PSD measurements, the most simple *Rectangular* window can be used, and mostly the highest frequency span and a resolution of 3200 lines is a proper setting.

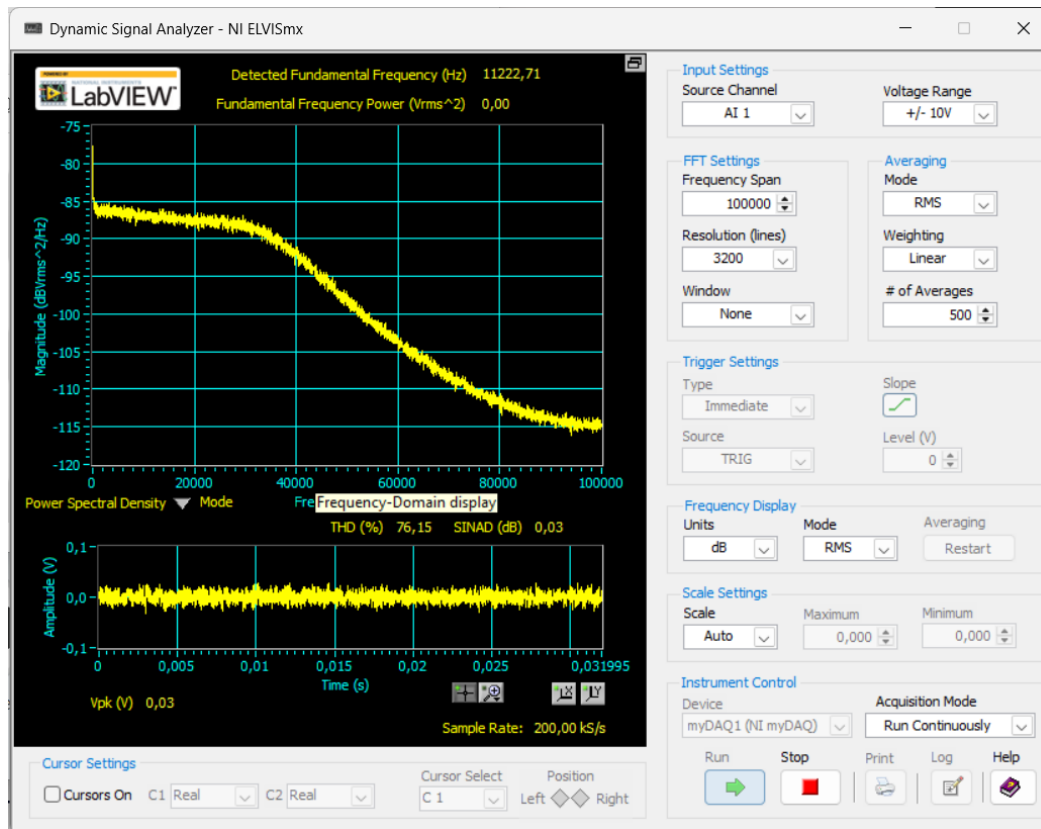


Figure II.3.1. NI ELVIS Dynamic Signal Analyzer program interface with settings well-suited for most of the measurements.

- In the **Averaging** menu you can set the averaging method and the weighting of the averaged results. The *Mode* is set to rms and the *Weighting* is set to linear. Also here you can set the number of averaged curves, which should be set to 500 for PSD measurements with 100kHz frequency span. When reducing the frequency window, the sampling frequency decreases, thus the measurement time increases, so it is reasonable to reduce the number of averaged curves (do not go below 30 averages).
- In the **Frequency Display** menu you can set the way to view the resulting spectrum, here we recommend using the RMS mode.
- In the Acquisition mode drop-down bar of the **Instrument Control** menu, you can select whether the sampling should be continuous or stop after the set curve number.
- Sampling can be started with the **Run button** and stopped with the **Stop button**.
- In case of continuous sampling, after the set curve number is completed, the **Restart** button becomes available, at which point averaging is complete and the spectrum can be saved by pressing the **Stop** button. It is important to note that the measurement started with the **Run** button will add the previous data points to the measurement, i.e. continue averaging, while Restart will delete the previous data points and start averaging from the beginning. Therefore **new measurements must be started with the Restart button**.
- By clicking on the **Log** button, you can choose the file to which the program **saves** the results of the last completed measurement. Avoid selecting a file that already

exists, as this will cause the fresh measurement data to be written after the previous data, making it quite inconvenient to decrypt. The structure of the extracted txt files is as follows: up to about 2/3 of the file is the date, time and raw voltage signal, followed by the frequency and the PS or PSD calculated from the signal. The data is extracted in decimal format.

4 NI ELVIS DC Voltage Source

This tool is used to drive the circuit with a constant DC voltage in the measurement of shot noise and $1/f$ -type noise. Note that the output of the myDAQ would be too much noisy by itself for such a sensitive measurement, so we also use a low-pass filter in the measurement circuit to suppress the output noise. Fig. II.4.1 shows the user interface with straightforward settings.

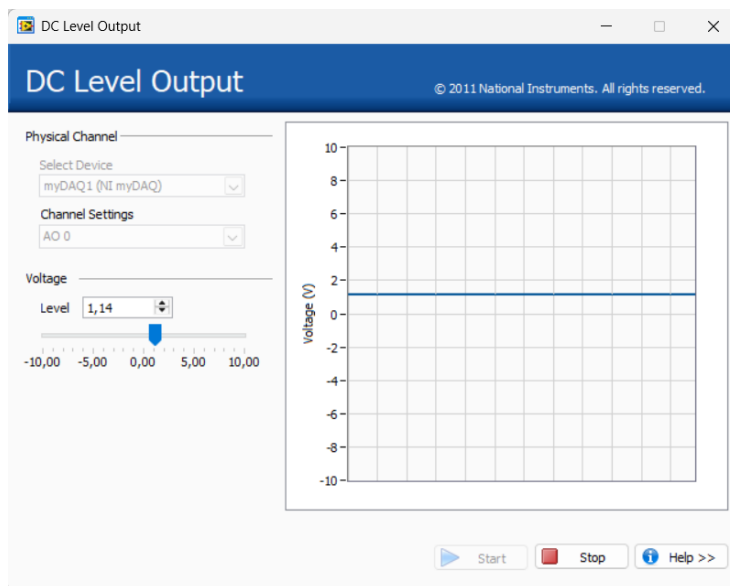


Figure II.4.1. User interface of the NI ELVIS DC level output tool.

5 NI ELVIS Function Generator and Oscilloscope

The function generator tool is used to output a triangular voltage sweep when the current-voltage ($I(V)$) curve of the diode is measured, while the response is recorded by the Oscilloscope tool.

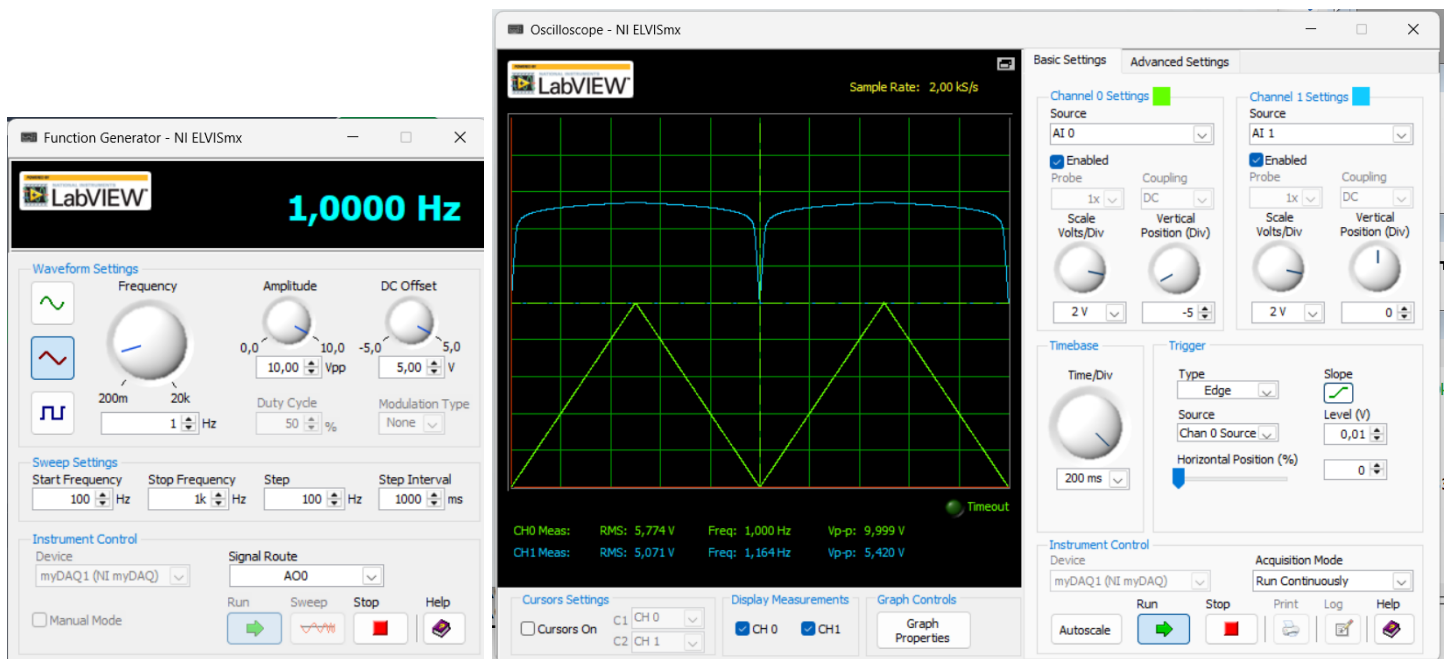


Figure II.5.1. User interface of the NI ELVIS Function generator (left) and Oscilloscope (right)

6 NI ELVIS Bode Analyzer

The Bode Analyzer tool is used to measure the transfer characteristic of the antialiasing filter by sweeping the driving frequency and measuring the filter response.

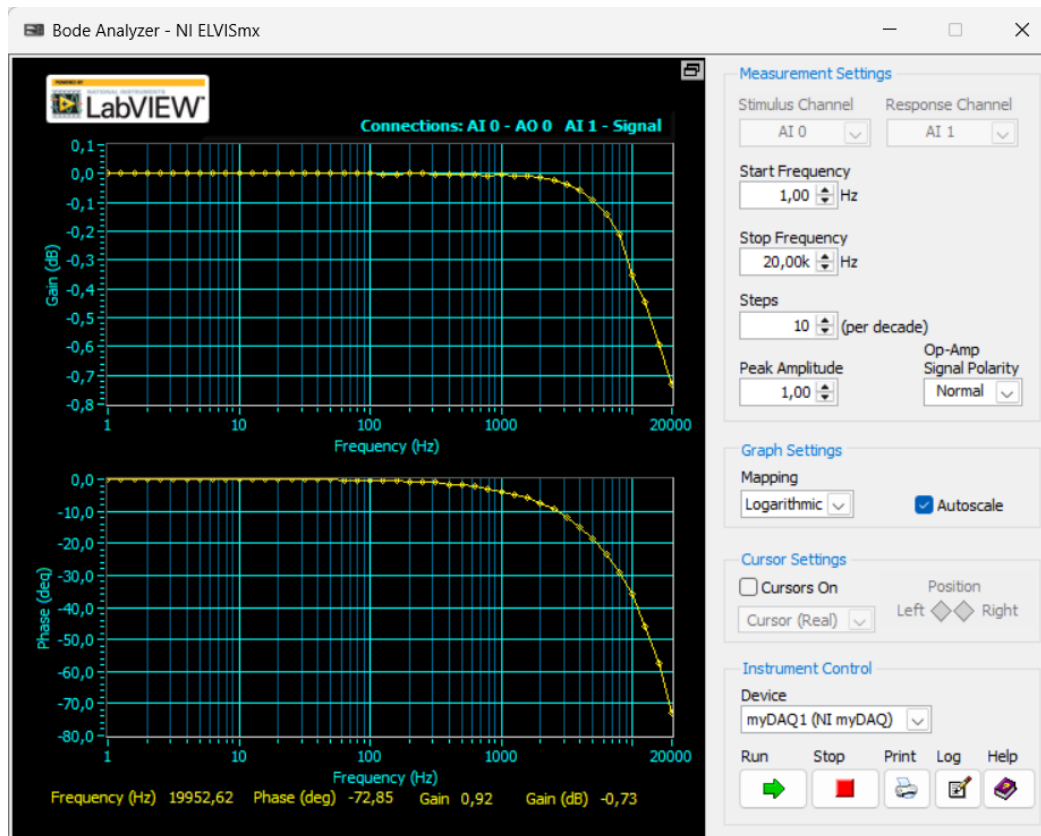


Figure II.6.1. User interface of the NI ELVIS Bode Analyzer tool.

7 NI ELVIS Digital Multimeter

The digital multimeter is used to accurately measure the resistances of the resistors used for the measurement.

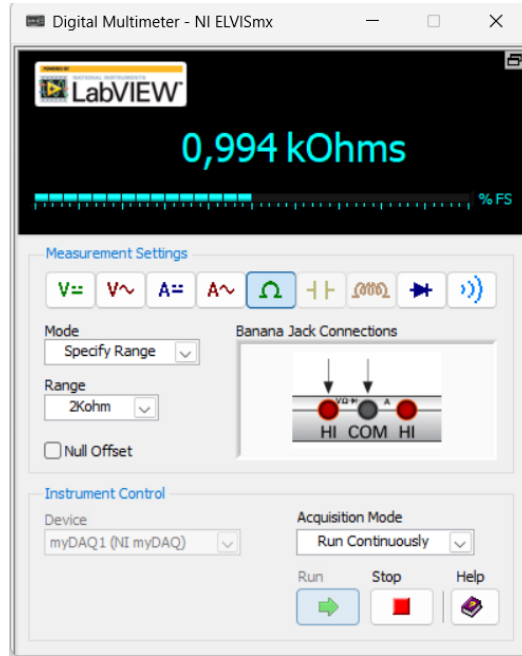


Figure II.7.1. User interface of the NI ELVIS Digital Multimeter.

8 FEMTO Low Noise Voltage Amplifier

For amplifying low noise levels, a Femto DLPVA 100-F-S or a Femto DLPVA 101-F-S amplifier is used. The control panel of the amplifier is shown in Fig. II.8.1.

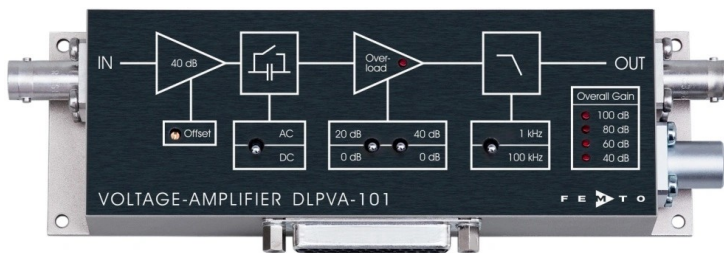


Figure II.8.1. Femto DLPVA 101-F-S low noise voltage amplifier

- The amplifier can be used in AC or DC coupling. In AC-coupled mode, a built-in filter cuts the DC component at 1.5 Hz.
- The basic amplifier gain is 20 dB, with an additional 60 dB (40 dB + 20 dB) gain optional. A gain of 80 dB is used for all noise measurement, while 20 dB for the diode $I(V)$ curve measurement.
- The frequency range can be selected from 1 kHz or 100 kHz. Above the selected frequency, a built-in low-pass filter cuts the amplified signal. For our measurements, use a bandwidth of 100 kHz.
- The input noise of the voltage amplifier is specified to be $\sqrt{S_0} \approx 5.5 \text{ nV}/\sqrt{\text{Hz}}$ for the FEMTO DLPVA-100-F-S and $\sqrt{S_0} \approx 5 \text{ nV}/\sqrt{\text{Hz}}$ for the FEMTO DLPVA-101-F-S at 80 dB gain. This means that when the amplifier input is short-circuited, instead

of the expected zero voltage, we see the amplifier's own voltage noise, which has a standard deviation of 5.5 nV for a 1 Hz bandwidth measurement.

- Always be careful not to overload the amplifier. This is indicated by a red LED light.

9 Antialiasing Filter Box

Since the 100 kHz bandwidth built into the amplifier is too high for the measurements to act as an antialiasing filter, an external third-order RLC low-pass filter in a sealable aluminium box with a cut-off at ≈ 30 kHz (see blue line in Fig. II.9.2) is installed to the output of the amplifier. Two BNC connectors on the side of the box provide the input and output. The filter circuit is illustrated in Fig. II.9.1.

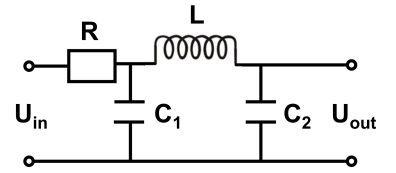


Figure II.9.1. Circuit diagram of the antialiasing filter. In the filter being used $C_1 = 2.2$ nF, $C_2 = 4.7$ nF, $L = 10$ mH and $R = 1500 \Omega$ values are applied.

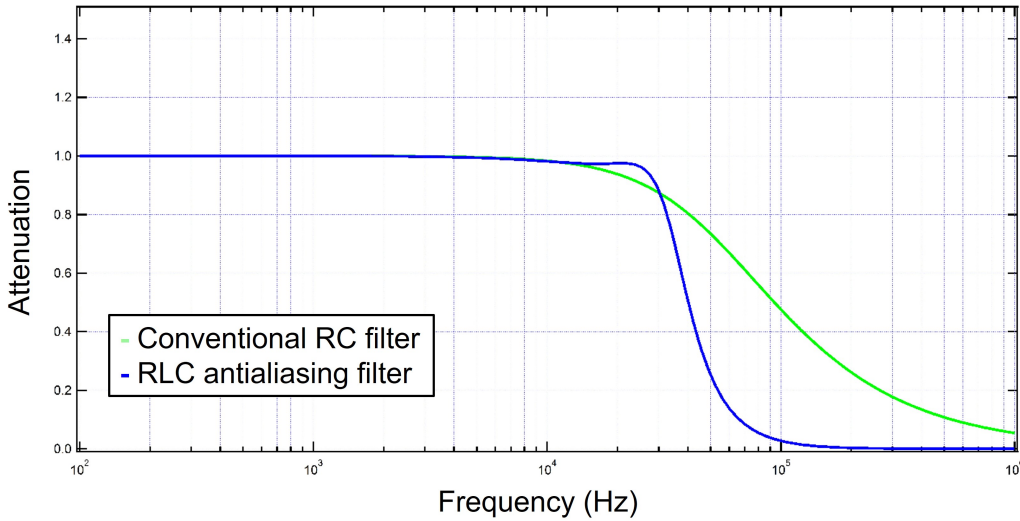


Figure II.9.2. Transfer characteristic of the antialiasing filter (blue line) in comparison to a conventional RC filter with similar cut-off.

10 Sample Holder Box

The circles for the measurement are assembled in a sealable metallic sample holder box using a zero insertion force socket, in which the pins can be closed by folding down a lever arm. See Fig. II.10.1 and its caption for details.

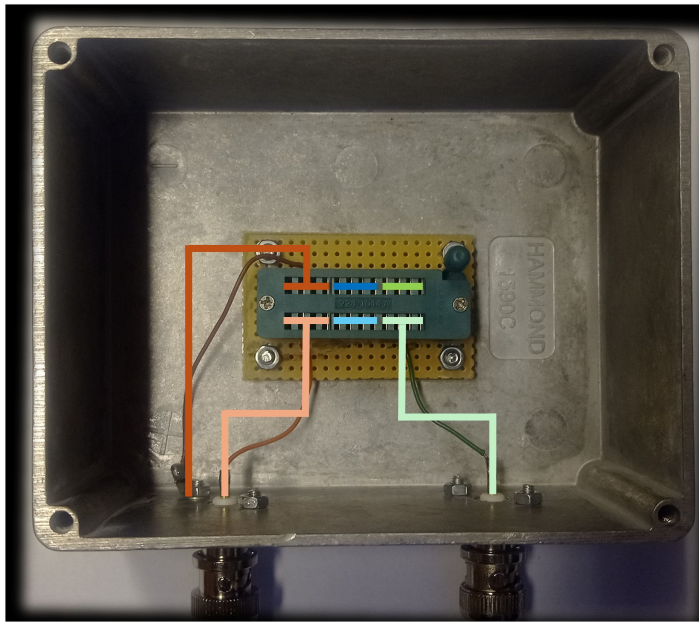


Figure II.10.1. Sample holder box with 2 front panel BNC connectors and a 2x12 zero insertion force (ZIF) sockets for assembling circuits. 4-4 adjacent holes of the ZIF socket are connected (see the colored lines with the different colors demonstrating the pins that are connected to each other). Two of the four-hole blocks are connected to the inner pin of 1-1 BNC connectors (light orange and green) and one to the sample holder box, i.e. the ground signal in the measurement (dark orange).

11 BD139 Bipolar Transistor

The shot noise measurement is performed on a semiconductor diode, which is practically an interface between a so-called p-doped and n-doped semiconductor layer. However, for accurate measurement, a device with sufficiently low levels of other noises (such as $1/f$ noise) is required. In practice, this is better achieved for noise-optimized transistors than for single diodes, so a BD139 bipolar NPN transistor (see Fig. II.11.1) is used for the measurement. Here, the investigated p-n junction is between the transistor base and emitter leads.

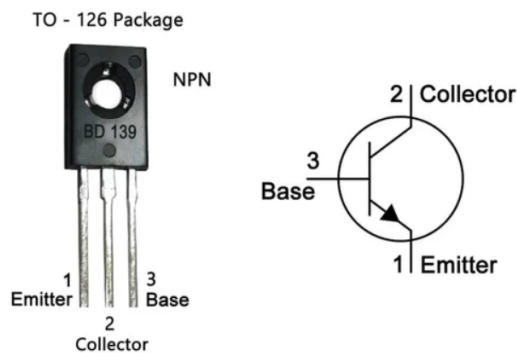


Figure II.11.1. Photo and circuit symbol of the BD139 NPN bipolar transistor. Between the base and the emitter contacts this transistor practically acts as a diode. [Source of figure and BD139 datasheet.](#)

12 Carbon Fibre Box

The carbon fibre used for the $1/f$ -type noise measurement is contained in a separate aluminium box. The box can be opened for inspection, but no action is required inside the box.

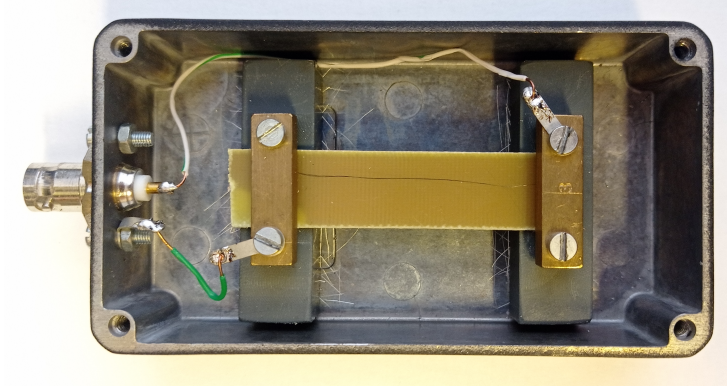


Figure II.12.1. Black painted box with a front panel BNC connector and clamps for fixing and electrically contacting the carbon fiber.

Measurement Tasks

III

Tasks Prior to the Measurement. It is very important that you come prepared for the measurement. In addition to reading the measurement description carefully, we ask you to be prepared for this session as follows:

1. The National Instruments ELVIS program package is essential for the measurement, make sure to have it installed on your computer prior to the first session. You may use the installer from this link: <http://ssplab.phy.bme.hu:5055/sharing/TZU3idyxp>.
2. It will make your lab work much easier if you come with a ready-made Matlab program that can convert the measured raw spectra to noise PSD measured in V^2/Hz . To do this, an example file saved by the dynamic signal analyzer is uploaded to the teams general channel folder. This contains the data of a measurement where a 50Ω terminated antialiasing filter is connected to the myDAQ input without the voltage amplifier. This practically shows the noise floor at the myDAQ input. The following remark and diagrams will help you to create this Matlab program.

Remark. The data displayed by the Dynamical Signal Analyzer should be converted from *decibel* to V^2/Hz or $V/\sqrt{\text{Hz}}$, and mostly replotted using logarithmic frequency scale and logarithmic noise scale.

The decibel scale is defined as follows. The amplification or attenuation in decibel is calculated as

$$A[\text{dB}] = 10 \cdot \log \left(\frac{P_{\text{out}}[\text{W}]}{P_{\text{in}}[\text{W}]} \right) = 20 \cdot \log \left(\frac{V_{\text{out}}[\text{V}]}{V_{\text{in}}[\text{V}]} \right).$$

The identity between the power-based and voltage-based formula relies on the quadratic scaling of the power with the voltage, $P \sim V^2$.

The unit dBV describes the ratio of the actual voltage and a reference voltage,

$$V[\text{dBV}] = 20 \cdot \log \left(\frac{V[\text{V}]}{V_{\text{ref}}} \right),$$

such that the reference voltage is $V_{\text{ref}} = 1 \text{ V}$.

Accordingly, the conversion of the PSD measured by the Dynamical Signal Analyzer in the unit of $\text{dBV}_{\text{rms}}^2/\text{Hz}$ converts to the unit of V^2/Hz according to the formula

$$S_V[\text{dBV}_{\text{rms}}^2/\text{Hz}] = 10 \cdot \log \left(\frac{S_V[\text{V}^2/\text{Hz}]}{S_{V,\text{ref}}} \right),$$

or

$$\frac{S_V[\text{V}^2/\text{Hz}]}{S_{V,\text{ref}}} = 10^{S_V[\text{dBV}_{\text{rms}}^2/\text{Hz}]/10},$$

where the reference noise level is $S_{V,\text{ref}} = 1 \text{ V}^2/\text{Hz}$.

The sample data of the Dynamical Signal Analyzer is illustrated in Fig. III.1.1a using

the units as the spectrum is plotted in the software ($S_V[\text{dBV}_{\text{rms}}^2/\text{Hz}]$ with linear axis vs. $f[\text{Hz}]$ with linear axis). This is converted to $S_V[\text{V}^2/\text{Hz}]$ and plotted with logarithmic noise and logarithmic frequency axis in Fig. III.1.1b. In engineering practice the noise of an instrument (amplifier, DAQ card, etc.) is often given in the units of $\text{V}/\sqrt{\text{Hz}}$. The last figure (Fig. III.1.1c) follows this engineering practice, where $\sqrt{S_V}[\text{V}/\sqrt{\text{Hz}}]$ with logarithmic axis vs. $f[\text{Hz}]$ with logarithmic axis is plotted. On the other hand S_V in the unit of $[\text{V}^2/\text{Hz}]$ is the primary quantity of noise, this is included in the noise formulae (Eqs. I.2.4, I.2.5), and S_V is additive for noise signals of different origin, while $\sqrt{S_V}$ is not additive. Later, we specify which plotting method is best suited to the specific task.

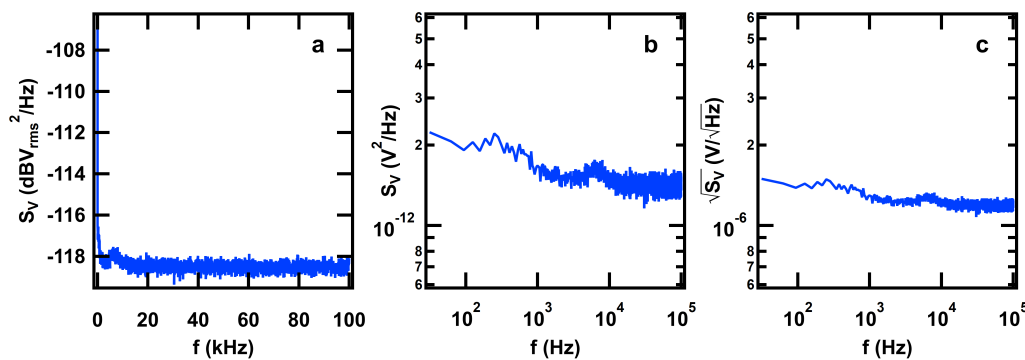


Figure III.1.1. Noise level at the myDAQ input plotted in different ways: a) $S_V[\text{dBV}_{\text{rms}}^2/\text{Hz}]$ with linear axis vs. $f[\text{Hz}]$ with linear axis, b) $S_V[\text{V}^2/\text{Hz}]$ vs. $f[\text{Hz}]$ with logarithmic noise and logarithmic frequency axis, c) $\sqrt{S_V}[\text{V}/\sqrt{\text{Hz}}]$ vs. $f[\text{Hz}]$ with logarithmic noise and logarithmic frequency axis.

Milestone. This measurement includes *milestones*. These are the most important outcomes, and also the key checkpoints where it is essential that the lab leader approves the results.

Task 1. Investigate the input noise of the voltage amplifier!

Power up the FEMTO amplifier, and set it to 80dB gain, 100kHz bandwidth and AC coupling. Terminate its input by 50Ω cap, which practically shortcircuits the input in this case. Connect the amplifier output to the antialiasing filter input and the filter output to the AI1± pins of the myDAQ Dynamic Signal Analyzer. Also connect AI1– to AGND. Check the input noise spectrum and compare it to the device specifications. Apply the recommended setting of the Dynamic Signal Analyzer (see Fig. II.3.1). What is the general input white noise level, and below what frequency do we see a higher 1/f-type noise level?

Hint for the Lab Notes. Plot the resulting noise spectrum on logarithmic-logarithmic scale and $\text{V}/\sqrt{\text{Hz}}$ unit following the scheme in Fig. III.1.1c. Do not forget to divide the spectrum by the gain of the amplifier. For comparison, plot the specified input noise level of the amplifier as a horizontal line in the same figure.

Milestone. The amplifier input noise is determined, and compared to the device specifications. This milestone result must be approved by the lab leader

before proceeding to the next tasks.

Hint for the Lab Report. Include the figure on the comparison of the measured input noise and the expected input noise in the lab report. Briefly summarize the method of the measurement, make sure the reader understands all notations of the figure, and discuss the figure, in particular the observed low and high frequency deviations from the expected input noise.

Task 2. Investigate the role of the antialiasing filter!

Repeat the previous exercise by excluding the antialiasing filter from the measurement system, and see how much error this change causes in the determination of the amplifier input noise level. Note, that in this case the error due to the aliasing phenomenon is moderate, as the FEMTO amplifier has an internal cutoff right at the 100 kHz Nyquist frequency of the measurement, i.e. aliasing only happens through the decaying tail of the FEMTO's internal filter.

To look even more clearly at the aliasing phenomenon put the antialiasing filter back to the measurement, but artificially reduce the Nyquist frequency of the sampling (i.e. set the frequency span to 30 kHz). Repeat the amplifier input noise measurement with this setting.

Hint for the Lab Notes. Include the raw spectra in the labnote.

Hint for the Lab Report. Prepare a figure comparing the three cases (100 kHz frequency span with and without antialiasing filter, and artificially reduced frequency span with antialiasing filter). Evaluate the error caused by aliasing and examine the foldback of the higher frequency components to below the Nyquist frequency for the case of artificially reduced Nyquist frequency. As the folding happens along the linear frequency scale, in this one case it makes sense to use linear noise and a linear frequency axis for the plot. Discuss the data.

Task 3. Refine the transfer characteristic of the antialiasing filter in the low-frequency domain!

From the above input noise measurement of the FEMTO amplifier the overall transfer characteristic of the antialiasing filter can be quite well seen. However, to see under what frequency a sufficiently small (e.g. less than 1 %) cut-off of the filter can be guaranteed, it is worth making a more accurate measurement with a Bode Analyzer.

Connect the myDAQ output (AO0 and AGND pins) to the input of the antialiasing filter box, and the filter output to the myDAQ input (AI1± pins). Measure back the output voltage by connecting AO0 to AI0+ and AGND to AI0−. Perform a frequency sweep with the Bode Analyzer. Note that due to the maximum sampling frequency of 200kS/s, this sweep only works up to a stop frequency of 20 kHz. Read the maximum frequency, below which the attenuation of the antialiasing filter causes less than 1 % error.

Hint for the Lab Notes. Plot the transfer characteristic (ratio of output voltage amplitude and input voltage amplitude as a function of frequency), and identify the corner frequency below which the attenuation is less than 1%.

Hint for the Lab Report. This is more of a technical detail, so if this measurement is well documented in the lab notes, a very brief description of the task, the specified corner frequency and a reference to the lab note is sufficient in the lab report.

Task 4. Determine the Boltzmann constant by thermal noise measurements!

Measurement Principle. When measuring thermal noise, the resistor (R) to be investigated is directly connected to the amplifier (see Fig. III.5.1). For proper shielding, however, it is mounted in the aluminum sample holder box with a BNC connector, between the inner point of the BNC connector and the ground point. The BNC connector is connected directly to the input of the amplifier using a BNC plug – BNC plug adapter. Since the measured thermal noise is comparable to the input noise of the amplifier (S_0), the measured noise will be the sum of the thermal noise and the input noise of the amplifier:

$$S_V = 4k_B T \cdot R + S_0.$$

Knowing the T ambient temperature and repeating the measurement at several R resistance values, the Boltzmann constant can be determined from the slope of the fitting line, while the y axis intersect gives the S_0 amplifier input noise.

Note, that the AI1± myDAQ pins are ground independent, but the metal shielding of the measurement system must be grounded. It is recommended to establish grounding with another coaxial cable, in which the cable shielding connects the metal sample box to the AGND (see Figure III.5.1). In this case, the inner conductor of the coaxial cable is not used. Furthermore, since the sample box is grounded by this coaxial cable, an additional grounding may cause a ground loop, i.e. the AI1– should not be connected to the AGND at the myDAQ terminal.

Set the FEMTO amplifier and the Dynamic Signal Analyzer like in the amplifier input noise measurement. Measure the voltage noise of different resistors, then determine the Boltzmann constant and the amplifier input noise by fitting. Use resistors of 1 kΩ, 3.3 kΩ, 6.8 kΩ and 10 kΩ for the measurement. During the measurement, calculate the expected noise density value for a given resistor and compare it to the measured value.

From the measured spectra, only those regions should be considered, which are clearly indicative of thermal noise with white noise spectrum. At low frequencies, this may be deviated by $1/f$ noise, and at high frequencies by filter attenuation. Peaks may appear in the spectrum at frequencies typical of the instrumentation in the environment and should be excluded when calculating the average noise density. The typical range for evaluating white noise is about 300 – 3000 Hz.

Hint for the Lab Notes. Plot all the spectra on the same figure using logarithmic frequency scale and logarithmic noise scale (like Fig. III.1.1b). Indicate the expected noise levels as horizontal lines (sum of thermal noise and amplifier input noise). Mind the units, and the gain of the amplifier! At least roughly read the S_V noise levels in the above 300 – 3000 Hz frequency window, and plot it as a function of the R resistance. Determine the Boltzmann constant from linear fitting.

Milestone. The Boltzmann constant is evaluated from the measured data and compared with the literature value. This milestone result must be approved

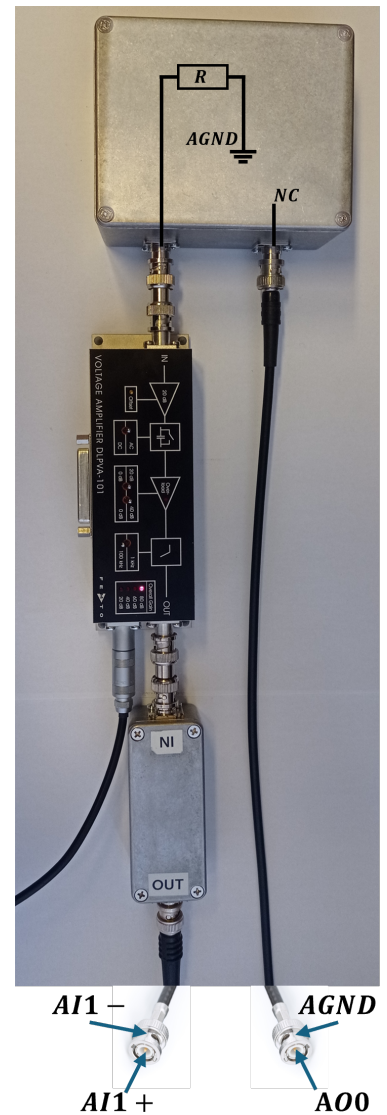


Figure III.5.1. Scheme of the thermal noise measurement. Note, that grounding is established through the coaxial cable to the right, AI1– should not be connected to the AGND at the myDAQ terminal.

by the lab leader before proceeding to the next tasks.

Hint for the Lab Report. Perform a more refined evaluation of the Boltzmann constant, if needed remove the pick-up peaks from the spectra. Summarize the results with two figures. (i) Plot all measured spectra on the same figure compared to the expected noise levels. Use legends to denote the resistance values corresponding to the different spectra. (ii) Include the S_V vs. R plot with the linear fit. Briefly summarize the task and results and make sure the reader understands all the notations in the figure. Discuss the results.

Optional Task. Investigate the role of the shielding for the measurement! Repeat the thermal noise measurement on the largest resistor in different arrangements, like opening the cover of the sample holder box, inserting unshielded cables like antennas to the circuit, placing 230 V instrument cables or switching mode power supplies close to the open sample holder box, etc.

Task 5. Measure the $I(V)$ curve of the diode, and determine its differential resistance to be used in the subsequent shot noise measurement!

Measurement Principle. Assemble the circuit in Fig. III.7.1 omitting the red capacitor. This is a current generator circuit that is driven by the AO0 myDAQ output, and the driving voltage is converted to the circuit current by the dominant $R_S = 330 \text{ k}\Omega$ resistor. The voltage drop on the diode (i.e. between the base and emitter contacts of the BD 139) is measured at the AI1 myDAQ input through the amplifier and the filter. The current is dominated by the ratio of the V_{AO0} drive voltage and the R_S resistance, as the voltage drop on the diode and on the $R_F = 330 \Omega$ resistor is negligible compared to the voltage drop on the large R_S resistor in series. (R_F will be applied later for filtering, but it is more convenient to have it in the circuit from the beginning.) Nevertheless the current through the diode can be accurately determined, if the V_{AO0} driving voltage is measured back at the AI0 myDAQ input:

$$I = \frac{V_{AO0} - V_{AI1}/A}{R_S + R_F} = \frac{V_{AI0} - V_{AI1}/A}{R_S + R_F}.$$

Here A is the amplification of the FEMTO.

After measuring the $I(V)$ curve it can be fitted with the common

$$I = I_0 (e^{\beta V} - 1)$$

exponential characteristic of a diode. This fitting can be traced back to linear fit in the $e^{\beta V} \gg 1$ limit (which will be highly satisfied in our subsequent measurements) after taking the logarithm of both sides:

$$\ln(I) = \beta V + \ln(I_0),$$

from which the differential resistance (see Fig. I.1.4) can be calculated as

$$R_d = \frac{1}{\beta \cdot I}.$$

Apply 20 dB FEMTO amplification in DC coupling. Apply a 10 V peak-to-peak trian-

gular signal with 5 V offset and 1 Hz frequency from the Function Generator to drive the diode for the $I(V)$ measurement. Measure back both the driving voltage and the voltage drop on the diode by the Oscilloscope. Find proper triggering and time-base settings, and save the data to the log file.

Plot the $I(V)$ curve and the $\ln(I)$ vs. V relation in MATLAB. Identify the $\approx 1 \mu\text{A}$ - $20 \mu\text{A}$ typical current range of the subsequent shot noise measurement and check that the linear $\ln(I)$ vs. V relation is already satisfied in this current range. Determine the β parameter by linear fitting to the $\ln(I)$ vs. V data in this current range.

Milestone. The β value of the diode is evaluated from the measured data. This milestone result must be approved by the lab leader before proceeding to the next tasks.

Hint for the Lab Report. Summarize your results with the $\ln(I)$ vs. V plot including the linear fit to the proper voltage region. Briefly summarize the task and the results and make sure the reader understands all the notations in the figure. Discuss the results.

Task 6. Determine the electron charge!

Measurement Principle. The scheme of the shot noise measurement is illustrated in Fig. III.7.1. The circuit is driven by the AO0 analog output of the myDAQ, which is filtered by a low-pass filter with very low cutoff frequency ($R_F = 330 \Omega$ and $C_F = 2200 \mu\text{F}$). The filtered output is fed through a resistor in series ($R_S = 330 k\Omega$, which defines the current in the circuit, and thereby realizes a current generator arrangement for the BD 139 bipolar transistor, which acts as a diode between its base and emitter connections. The emitter (E) is grounded, while the $S_V(f)$ voltage noise spectrum is measured at the base (B) through the FEMTO amplifier and the antialiasing filter. According to Eq. I.2.4 the current noise of the diode is $S_I(f) = 2eI$. The I current at a certain $V_{A00} = V_{AI0}$ driving voltage is known from the previous $I(V)$ curve measurement, i.e. the proper driving voltages can be matched to the desired current values. We have also seen, that the S_I current noise of a nonlinear device converts to voltage noise as $S_V = R_d^2 \cdot S_I$, where R_d is the differential resistance of the diode at the operation point, i.e. at the set I current. This differential resistance was also evaluated in the previous task. Based on all these the voltage noise spectrum of the diode can be written as:

$$S_V = S_I \cdot R_d^2 + S_0 = 2eI \cdot R_d^2 + S_0,$$

where S_0 is the base noise level of the amplifier determined previously.

Applying various voltage levels at the AO0 output of the myDAQ one can tune the operating current of the diode. Plotting the S_V voltage noise level against IR_d^2 and fitting a line to the data the slope gives twice the electron charge, while the y axis intersect is the S_0 base noise level of the amplifier.

Change the circuit of the previous task by inserting the filtering capacitor, but not changing anything else.

Remark. Note, that the filter with the capacitor introduces very long transients, i.e. always wait at least 10 s prior to recording the spectra once you change the driving

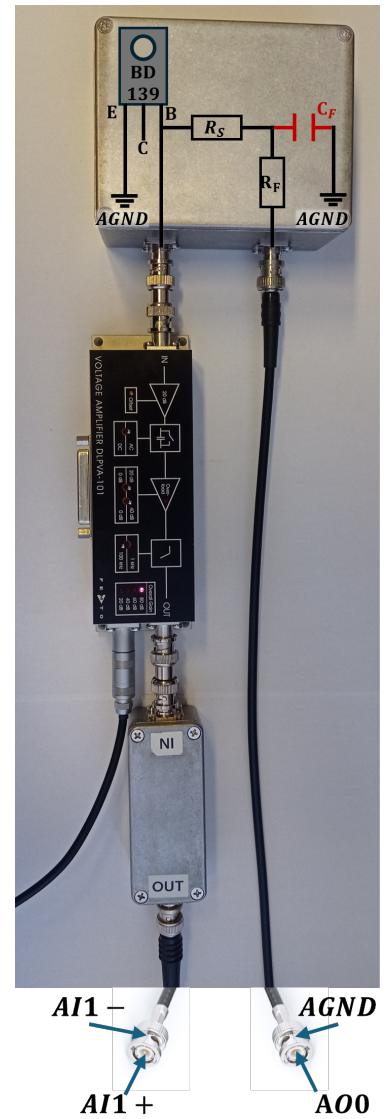


Figure III.7.1. Scheme of the shot noise measurement. Note, that A00 should be measured back at the AI0 input, i.e. connect A00 to AI0+ and AGND to AI0- at the myDAQ terminal. Mind the polarity, when inserting the C_F electrolyte capacitor!

voltage. You may calculate the RC time-constant of the filter.

Set the FEMTO amplifier and the Dynamic Signal Analyzer like in the amplifier input noise measurement. From the previous task, read the V_{AO0} driving voltage values, at which currents close to $2\mu\text{A}$, $3\mu\text{A}$, $4\mu\text{A}$, $5\mu\text{A}$, $7\mu\text{A}$ and $10\mu\text{A}$ values are realized, and record the PSD spectra at these settings. Determine the electron charge according to the above measurement principle. Use the results of the previous task to (i) assign the accurate current values to the set V_{AO0} driving voltage values and (ii) to insert the $R_d = 1/(\beta I)$ differential resistance to the above analysis. Use the average PSD values in the $300\text{ Hz} - 3\text{ kHz}$ frequency range for the evaluation.

Hint for the Lab Notes. Put all the recorded noise vs. frequency data on the same figure following the plotting style in Fig. III.1.1b. Calculate the expected S_V values considering the base amplifier noise level, the current and the differential resistance, and plot these as horizontal lines on the same figure. Compare the results with the expectations. At least roughly read the S_V values in the $300\text{ Hz} - 3\text{ kHz}$ range and plot these as a function of IR_d^2 . Determine the electron charge from the slope. This is a complicated evaluation, which you should always carry out during the measurement to recognize if the result is incorrect and, in the latter case, to find the error.

Milestone. The electron charge is evaluated from the measured data and compared with the literature value. This milestone result must be approved by the lab leader before proceeding to the next tasks. Missing the last measurement task is less of an issue than incorrectly completing this milestone.

Hint for the Lab Report. If needed, perform a more refined evaluation of the electron charge e.g. by removing the pick-up peaks from the spectra. Summarize the results with two figures. (i) Plot all measured spectra on the same figure compared to the expected noise levels. Use legends to denote the corresponding current and differential resistance values. (ii) Include the S_V vs. IR_d^2 plot with the linear fit. Briefly summarize the task and results and make sure the reader understands all the notations in the figure. Discuss the results.

Task 7. Investigate the $1/f$ -type noise of a carbon fibre!

Measurement Principle. $1/f$ -type noise is investigated on a carbon fibre. This is a circuit element with highly linear $I(V)$ curve, i.e. $I(V)$ characterization is not necessary, but prior to the measurement its resistance should be determined with the Digital Multimeter. Afterwards the PSD spectra are investigated at various current values likewise in the previous task, but applying larger currents, and therefore smaller $R_S = 33\text{ k}\Omega$ resistor in series. Here, the specialty is, that $S_V = C \cdot (f/f_0)^{-\gamma}$ spectra are expected with $\gamma \approx 1$ instead of the white noise spectrum of the thermal noise and shot noise. Here, C is a constant describing the noise amplitude, and f_0 is a reference frequency, e.g. $f_0 = 1\text{ Hz}$. Furthermore, according Eq. 1.2.6, $S_V \sim I^2$ scaling is expected instead of the linear current dependence of the shot noise power, which yields $C \sim I^2$. These two characteristic properties of the $1/f$ -type noise are investigated here using the measurement scheme in Fig. III.8.1.

Apply various V_{AO0} driving voltage values in the $0 - 10\text{ V}$ range, and determine the



Figure III.8.1. Scheme of the $1/f$ -type noise measurement. Here, $R_S = 33\text{ k}\Omega$ resistor in series is applied. Mind the polarity, when inserting the C_F electrolyte capacitor!

corresponding currents. Record the PSD spectra at these settings. Use the spectrum at zero voltage as a baseline (i.e. the thermal noise of the carbon fibre), which should be subtracted from the finite voltage spectra.

Determine the γ power coefficient by fitting the frequency-dependence of the spectra. For that calculate the $\log(S_V)$ vs. $\log(f/f_0)$ curve and apply a linear fit to that according to the $\log(S_V) = -\gamma \log(f/f_0) + C$ relation, where C is a constant and $f_0 = 1$ Hz. Note, that if the data are fitted by a line on the log-log plot, it is recommended to resample the data for about 10-20 points equidistantly spaced along the logarithmic frequency axis, otherwise the high frequency components would dominate the fit.

Plot the spectra by normalizing them to I^2 , and check, whether these normalized spectra fall on the same curve, i.e. whether the I^2 current dependence is justified.

Hint for the Lab Notes. Plot the measured spectra in the labnote.

Hint for the Lab Report. Evaluate the frequency and voltage dependence of $1/f$ noise in the lab report. Summarize the results with three figures. (i) Plot all measured spectra in the same figure. Use legends to denote the corresponding current values. (ii) Select one of the spectra and perform the linear fit of $\log(S_V)$ vs. $\log(f/f_0)$. Note that in (i) we only used logarithmic axes, but for the fit we actually need to calculate the logarithms of the data. (iii) Re-plot figure (i) by normalizing the curves to I^2 and check that the quadratic current dependence is satisfied. Briefly summarize the task and results and make sure that the reader understands all the notations in the figures. Discuss the results.